
Improving NLP Classification Models Through Discourse Marker Pre-training

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Abstract. Pre-trained Language Models have achieved state-of-the-art results across various Natural Language Processing (NLP) tasks. However, their ability to capture complex inter-sentence relationships remains an area for improvement. This work investigates whether pre-training models to predict Discourse Markers (DMs)—explicit linguistic signals such as "however" or "therefore"—can enhance performance on downstream classification tasks. We constructed multilingual DM datasets for English, Italian, and Portuguese using a self-supervised mining approach on Wikipedia. We then evaluated the impact of DM supervision using both Transfer Learning and Multi-Task Learning (MTL) strategies across four downstream tasks: Natural Language Inference (NLI), Semantic Textual Similarity (STS), Hate Speech Detection, and Topic Classification. Our experiments show that DM-enhanced models outperformed baselines in 75% of the cases, with the most significant gains observed in semantic tasks (STS) and lower-resource languages (Italian). The results suggest that while discourse information is highly beneficial for tasks requiring logical reasoning, its application must be carefully tuned to avoid negative transfer in unrelated tasks.

Keywords. NLP, LLM, Discourse Markers

1 Introduction

The advent of Transformer-based models, such as BERT and RoBERTa, has revolutionized the field of Natural Language Processing. These models learn rich contextual representations by training on massive text corpora. Despite their success, standard pre-training objectives (such as Masked Language Modeling) do not always explicitly capture the logical and coherent relationships between consecutive sentences.

Discourse Markers (DMs) are lexical cues that explicitly signal the semantic relation between textual segments (e.g., contrast, elaboration, or implication). Because DMs are naturally abundant in text, they offer a low-cost, self-supervised signal that can be leveraged to teach models about discourse structure without manual annotation.

In this report, we propose a pipeline to mine and categorize discourse markers in three languages: English, Portuguese, and Italian. We aim to answer two primary research questions: (1) Does incorporating discourse marker objectives improve performance on downstream tasks? and (2) Does the effectiveness of this approach vary depending on the linguistic resources available or the nature of the task (e.g., NLI-oriented vs. general classification)?

We compare three experimental settings: a standard baseline, a sequential Transfer Learning approach, and a Multi-Task Learning (MTL) framework. Our findings indicate that discourse knowledge is particularly effective for tasks involving sentence-pair reasoning and can serve as a regularizer for languages with limited pre-training resources.

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2 Discourse Markers Dataset

2.1 Data Mining Methodology

To construct the multilingual discourse marker (DM) datasets, we implemented a self-supervised mining pipeline inspired by the methodology proposed in the Discovery dataset [4]. The source data was obtained from the `wikimedia/wikipedia` dump (20231101 version) via the Hugging Face library, using language-specific splits for English, Italian, and Portuguese.

The raw text was processed using the `spaCy` library (specifically the `{lang}_core_news_sm` models) to perform sentence segmentation. We designed a heuristic algorithm to identify pairs of adjacent sentences, denoted as (S_1, S_2) , that exhibit a discursive relationship. A valid pair is defined by the following criteria:

1. The first sentence (S_1) must contain a valid syntactic termination (ending with a period, semicolon, exclamation point, or question mark).
2. The second sentence (S_2) must begin with a known discourse marker followed immediately by a comma.
3. To ensure the semantic quality of the samples and avoid tokenization artifacts, both sentences are filtered based on length constraints, requiring a token count $3 \leq L \leq 32$.

We also removed the discourse markers from S_2 . This prevents the model from relying on the explicit presence of the marker, forcing it instead to learn the underlying semantic relationship between the sentences to predict the correct label.

2.2 Taxonomy and Class Aggregation

The definition of valid discourse markers and their corresponding classes varied by language due to resource availability:

- **Italian:** Markers and initial classifications were provided by the university.
- **English:** Markers were adopted from the taxonomy provided in the Discovery repository².
- **Portuguese:** Markers were sourced from the SciPo project³.

Given that the data was mined from the wild (Wikipedia), the natural frequency of discourse markers resulted in a highly imbalanced distribution. To mitigate this and facilitate robust model convergence, we aggregated fine-grained markers into broader semantic categories using Large Language Models (LLMs). As the primary objective of this work is to leverage DMs for transfer learning in downstream NLP tasks rather than pure linguistic analysis, a high-granularity classification was deemed unnecessary.

The resulting taxonomy consists of the following high-level classes: **Contrastive** (CDM), **Elaborative** (EDM), **Implicative** (IDM), and **Temporal** (TDM). For languages with insufficient samples in specific categories (specifically Italian), IDM and TDM were merged into a single **Sequential** (SEQ) class.

2.3 Data Distribution

Table 1 presents the final distribution of samples per class for each language after the mining and filtering process.

²<https://github.com/sileod/Discovery>

³<https://escritacientifica.sc.usp.br/scipo/>

Tabela 1: Distribution of extracted Discourse Marker pairs by language and semantic class.

Language	Class	Count
English (en)	Contrastive (CDM)	183,908
	Elaborative (EDM)	109,061
	Temporal (TDM)	101,306
	Implicative (IDM)	43,071
	<i>Total</i>	437,346
Portuguese (pt)	Contrastive (CDM)	91,449
	Elaborative (EDM)	50,459
	Implicative (IDM)	39,982
	<i>Total</i>	181,890
Italian (it)	Contrastive (CDM)	50,582
	Elaborative (EDM)	18,694
	Sequential (SEQ)	6,462
	<i>Total</i>	75,738

3 Tasks and Datasets

To evaluate the impact of Discourse Marker (DM) pre-training, we selected four distinct NLP tasks divided into two categories: *NLI-oriented tasks* (Natural Language Inference and Semantic Textual Similarity), which share structural similarities with the discourse marker objective, and *General Classification tasks* (Hate Speech Detection and Topic Classification), which represent distinct downstream applications.

For the NLI-oriented tasks, we utilized multilingual datasets to facilitate cross-lingual comparisons. For the General Classification tasks, we selected language-specific benchmarks for English, Italian, and Portuguese.

3.1 NLI-Oriented Tasks

3.1.1 Natural Language Inference (NLI)

We utilized the `multilingual-NLI-26lang-2mil7` dataset⁴. This corpus comprises machine-translated versions of five major NLI benchmarks: MultiNLI, Fever-NLI, ANLI, LingNLI, and WANLI. The aggregated dataset consists of approximately 105k samples across 26 languages. Each sample contains a *premise*, a *hypothesis*, and a label classified as *entailment* (0), *neutral* (1), or *contradiction* (2). We combined all samples and performed a stratified random split, allocating 90% for training and 10% for testing.

3.1.2 Semantic Textual Similarity (STS)

We employed the machine-translated version of the STS Benchmark [3]. The original dataset provides pairs of sentences with a similarity score ranging from 0.0 to 5.0. To align this with our classification framework, we converted the regression target into a binary classification task. A pair is labeled as positive (1) if the similarity score $s \geq 3.0$, and negative (0) otherwise. The resulting dataset contains 5,750 samples for training and 1,380 for testing.

⁴<https://huggingface.co/datasets/MoritzLaurer/multilingual-NLI-26lang-2mil7>

3.2 General Classification Tasks

3.2.1 Hate Speech Detection

We selected binary classification datasets where texts are labeled as *hate speech* or *non-hate speech*:

- **English:** We used the *hate* subset of the `tweet_eval` benchmark⁵. The dataset consists of 9,000 training samples and 2,970 test samples.
- **Italian:** We utilized the development set (Tasks A and B) from the HaSpeeDe 2 shared task [1] at EVALITA 2020. The 6,800 samples were randomly split into 90% training and 10% testing.
- **Portuguese:** We employed HateBR [5], a balanced dataset of comments from social media. We utilized the binary classification version (7,000 rows) with a 90%/10% stratified split.

3.2.2 Topic Classification

We selected multi-class news categorization datasets for each language:

- **English:** We used the AG News dataset⁶, a balanced corpus classifying news into four categories: *World*, *Sports*, *Business*, and *Sci/Tech*. It contains 120k training samples and 7.6k test samples.
- **Italian:** We used the "massive" version of the ITALIC dataset [2], which focuses on intent classification. To ensure class stability, we filtered the dataset to include only scenario labels with at least 500 samples. The final subset consists of 10 distinct classes, with 8,808 samples for training and 2,303 for testing.
- **Portuguese:** We curated a dataset of news articles from the Brazilian newspaper *Folha de S.Paulo*⁷. We filtered the data to retain only categories with over 1,000 samples, resulting in 18 classes. The final dataset comprises 160k samples, divided into a stratified 90%/10% split.

4 Experimental Setup

4.1 Training Strategies

We evaluated the performance of the models across three languages and four tasks using three distinct training strategies:

1. **Baseline:** The downstream task is trained directly on the pre-trained base model without any DM exposure.
2. **Transfer Learning (Sequential):** The model is first fine-tuned on the Discourse Marker classification task. The weights are then frozen and further fine-tuned on the downstream task.
3. **Multi-Task Learning (MTL):** The model is trained simultaneously on both the downstream task and the Discourse Marker classification.

⁵https://huggingface.co/datasets/cardiffnlp/tweet_eval

⁶https://huggingface.co/datasets/wangrongsheng/ag_news

⁷<https://www.kaggle.com/datasets/marlesson/news-of-the-site-folhaol>

4.2 Multi-Task Implementation

In the MTL setting, the model optimizes a joint loss function defined as:

$$\mathcal{L}_{total} = \mathcal{L}_{task} + \beta \cdot \mathcal{L}_{dm} \quad (1)$$

where $\mathcal{L}_{task}$ is the downstream task loss, $\mathcal{L}_{dm}$ is the discourse marker loss, and β is a hyperparameter balancing the auxiliary task. We experimented with $\beta \in \{0.2, 0.4, 0.6, 0.8, 1.0\}$. The reported performance corresponds to the best β value found for each task.

To ensure balanced training, we implemented a custom batch sampler that constructs batches containing a 50/50 ratio of samples from the downstream task and the DM dataset. This allows the model to separate inputs for their respective classification heads efficiently.

4.3 Hyperparameters and Infrastructure

All experiments were conducted using the PyTorch framework on a university server equipped with high-performance GPUs. We employed the following pre-trained transformer models available via Hugging Face:

- **English:** RoBERTa-base (`roberta-base`) ⁸.
- **Portuguese:** BERTimbau (`bert-base-portuguese-cased`) ⁹.
- **Italian:** UmBERTo (`umberto-commoncrawl-cased-v1`) ¹⁰.

Training was performed with a batch size of 64, a learning rate of 1×10^{-5} , and a weight decay of 0.01. The maximum sequence length was set to 512 tokens. Each run was capped at 30 epochs with an early stopping patience of 5 epochs. The source code and reproduction scripts are publicly available on GitHub¹¹.

5 Results

The quantitative results of our experiments are summarized in Table 2. We report the F1-score for all tasks across the three languages. The best performance for each language-task combination is highlighted in bold.

As detailed in Table 2, incorporating Discourse Markers (DM) improved the model performance in 9 out of 12 experiments (75%). However, the optimal training strategy varies significantly depending on the nature of the downstream task.

For **NLI-like tasks**, both Transfer Learning and Multi-Task Learning (MTL) proved effective. Notably, the Semantic Textual Similarity (STS) task showed the most consistent and significant gains across all languages (e.g., +1.66% in Portuguese). This aligns with the intuition that discourse markers explicitly signal the semantic relationship between sentences (e.g., contrast, implication), which is the core objective of STS tasks.

Conversely, for **Non-NLI tasks**, Transfer Learning consistently underperformed, often degrading performance compared to the baseline (e.g., Hate Speech in English dropped from 0.5393 to 0.4940). This suggests that pre-training on DMs may shift the model’s weights too far towards

⁸<https://huggingface.co/FacebookAI/roberta-base>

⁹<https://huggingface.co/neuralmind/bert-base-portuguese-cased>

¹⁰<https://huggingface.co/Musixmatch/umberto-commoncrawl-cased-v1>

¹¹<https://github.com/GuiFigueiredo/DiscourseMarkers>

Tabela 2: Comparison of F1-scores across Baseline, Multi-Task Learning (MTL), and Transfer Learning settings. The best result for each row is in bold.

Type	Task	Language	F1-Score		
			Baseline	MTL	Transfer
NLI-Like	Natural Language Inference	EN	0.8027	0.7999	0.7991
		IT	0.7070	0.7067	0.7110
		PT	0.7245	0.7067	0.7256
	Semantic Textual Similarity	EN	0.8644	0.8716	0.8629
		IT	0.7876	0.7989	0.8013
		PT	0.8057	0.8223	0.8106
Non-NLI	Hate Speech Detection	EN	0.5393	0.5258	0.4940
		IT	0.7997	0.8002	0.7795
		PT	0.9257	0.9286	0.9186
	Topic Classification	EN	0.9491	0.9526	0.9491
		IT	0.8977	0.9031	0.8889
		PT	0.8956	0.8939	0.8907

discourse-level features, causing "negative transfer". In these cases, MTL proved to be the superior strategy, likely because the auxiliary DM loss acts as a regularizer rather than a conflicting objective.

We observed a correlation between the improvement margin and the resource richness of the language. In English, the most high-resource language with a robust base model (RoBERTa), the baseline was difficult to beat, maintaining the best score in 2 out of 4 tasks. In contrast, for Italian, which relies on a smaller pre-trained ecosystem, our methods improved upon the baseline in 100% of the cases. This suggests that injecting discourse knowledge is particularly beneficial for lower-resource languages where the base models may not have fully captured complex inter-sentence semantic relations during their initial pre-training.

Figure 1 visualizes the magnitude of these improvements. While the majority of experiments showed positive gains, only three cases exceeded a 1% absolute improvement. It is important to note that due to computational constraints, these results represent single experimental runs. Consequently, marginal differences should be interpreted with caution, as they may fall within the range of stochastic variation inherent to seed selection. Nevertheless, the consistent trend in STS and Italian tasks points to a genuine structural benefit of discourse marker supervision.

6 Conclusion

In this work, we explored the utility of Discourse Marker (DM) prediction as an auxiliary objective for improving NLP classification models. By mining data from Wikipedia, we created silver-standard datasets for English, Italian, and Portuguese and evaluated the impact of this additional supervision on four diverse downstream tasks.

Our results demonstrate that teaching models to recognize discourse relations yields performance improvements in the majority of experimental scenarios (9 out of 12). The benefits are most pronounced in *NLI-like* tasks, particularly Semantic Textual Similarity (STS), where understanding the logical connection between sentences is paramount. Furthermore, we observed a resource-dependent trend: while the highly optimized English baseline was difficult to beat, the

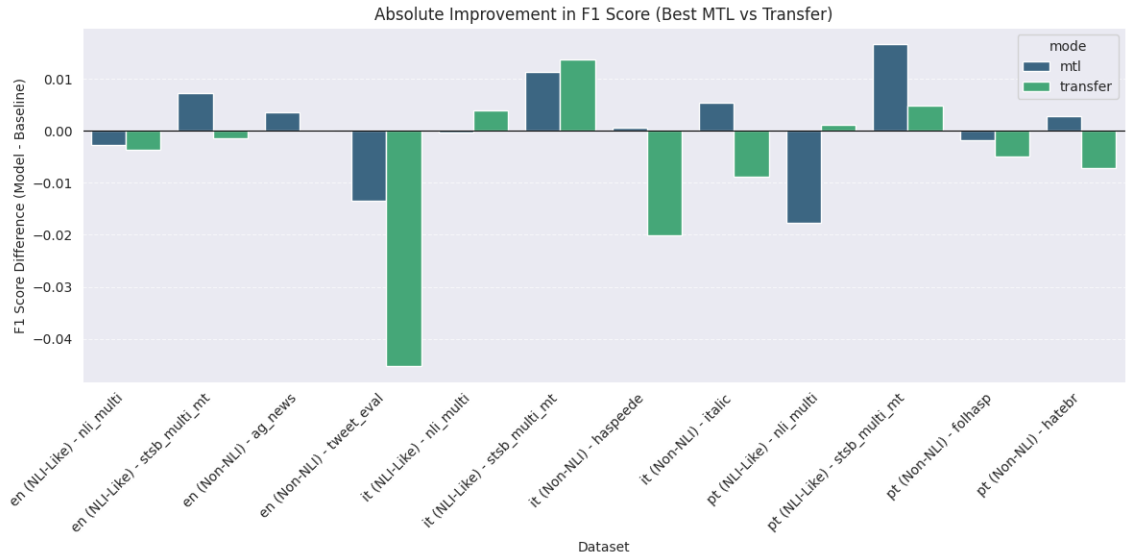


Figure 1: Absolute improvement in F1 Score compared to the baseline. Positive values indicate that either MTL or Transfer Learning outperformed the base model.

Italian models (representing a lower-resource setting) showed consistent improvements across all tasks.

However, we also identified the risk of negative transfer, particularly in *Non-NLI* tasks like Hate Speech Detection, where heavy pre-training on discourse features occasionally degraded performance. Multi-Task Learning proved to be a more robust strategy in these cases compared to sequential Transfer Learning.

Future work should focus on validating these findings with multiple random seeds to ensure statistical significance and exploring distinct weighting mechanisms to better balance the discourse objective with downstream tasks. Ultimately, this research highlights the potential of discourse markers as a computationally efficient signal to enhance the reasoning capabilities of multilingual language models.

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